# 潜在失效模式及后果分析（PFMEA）
# Potential Failure Mode and Effects Analysis

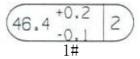
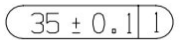
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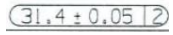
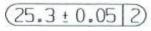
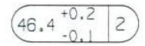
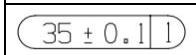
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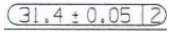
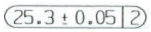
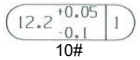
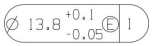
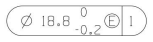
产品名称Part Name： 磁性罩　☐样件Sample　☐试生产Trial run　√生产Production　修订日期Revised date：24.8.26

核心小组Core Team:马生勇、 孙伟亮、陈金龙、程田峰、王彦军　编制人Made by：孙伟亮

| 产品过程编号Process No. | 项目/功能 Item/Function | 潜在失效模式 Potential failure mode | 潜在失效后果 Potential failure effects | 严重度 Severity S | 级别 Level | 潜在失效起因/机理 Potential cause/ Mechanism of failure | 频度 Occurrence O | 现行过程控制 Current Controls 预防 Prevent | 现行过程控制 Current Controls 探测 Detect | 探测度 Detectivity D | RPN | 建议措施 Recommended action | 责任及目标完成日期 Responsibility & Target completion | 措施结果 Actions results 采取的措施 Actions taken | S | O | D | RPN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 原材料入库检验 Raw material in-house inspection | 材料表面有锈斑、伤痕等 There is stain and scar etc. in the material | 无法使用，外观不能满足客户要求 Apperance can't meet customer's need, and it can't be used | 4 | | 材料未用包装纸包好/运输过程中变形导致伤痕 Material is not packed well, results in scar during delivery. | 3 | 供应商过程控制Supplier's process control | 加强进货检验和过程外观检验，增加进货抽检量 Reinforce in-house inspection and process apperance inspection. Increase the QTY of the samples for casual inspection | 6 | 72 | | | | | | | |
| | | 尺寸规格不符 The dimension Spec. is unconformity | 无法生产产品 Can't do production | 5 | | 供应商未按规定尺寸备料 Supplier doesn't prepair material according to specified dimension | 2 | 制订尺寸规格的公差标准 Make tolerance standard of dimension SPEC | 进货检验；合格供应商处采购 In-house inspection; purchase from qualified supplier | 6 | 60 | | | | | | | |
| | | 表面粗糙度不符 Surface roughness is unconformity | 无法生产产品 Can't do production | 5 | | 供应商未按规定尺寸备料 Supplier doesn't prepair material according to specified dimension | 2 | 制订尺寸规格的公差标准 Make tolerance standard of dimension SPEC | 进货检验；合格供应商处采购 In-house inspection; purchase from qualified supplier | 6 | 60 | | | | | | | |
| | | 表面硬度超差 Surface hardness deviation | 无法生产产品 Can't do production | 5 | | 供应商未按规定尺寸备料 Supplier doesn't prepair material according to specified dimension | 2 | 制订尺寸规格的公差标准 Make tolerance standard of dimension SPEC | 进货检验；合格供应商处采购 In-house inspection; purchase from qualified supplier | 6 | 36 | | | | | | | |
| | | 材料性能、成分等不合格 Material has unqualified performance and component | 无法进行正常生产 Can't do production nomally | 9 | D | 供应商自检失误/材料发错 Supplier do wrong self inspection, and material is wrong delivered | 1 | 供应商每批提供质保书 Supplier provides quality guarantee note of each batch | 认可时测试；定期委外作性能等测试 Do confirmation test,outsoucing for test the performance regually | 7 | 63 | 每年委托第三方检测 Entrust a third party testing per year | 程田峰 chen zhenhui | 第三方检测 A third party testing | 9 | 1 | 6 | 54 |
| 2 | 存储材料 Material Storage | 材料变形表面损伤 aterial is deformed and has scar on surface | 影响产品外观/材料变形无法使用 Affect part surface/material | 4 | | 摆放不合理造成倒塌，材料安放时无固定措施 Placing unreasonable lead the collapse, no fixed measure for material placement | 1 | 原材料安放时两边用木块固定 Raw material is fixed by wood block on both sides when placed | 仓库每月检查 Check the warehouse each | 7 | 28 | | | | | | | |
| | | 材料生锈 Material is rusty | 无法使用，外观不能满足客户要求 Apperance can't meet customer's need, and it can't be used | 4 | | 温湿度条件不符合要求 The temperature and wetness can't meet the need | 2 | 专用仓库 Special warehouse | 及时检查 Check on-time | 3 | 24 | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | S | | | O | 预防 Prevent | 探测 Detect | D | | | | 采取的措施 Actions taken | S | O | D | RPN |
| 3 | 模具冲压 Tooling stamping | 模具闭合高度不符 Tooling shut height is inconsistent | 产品不合格，送料不畅 Part is not qualified, feed is not smooth | 6 | | 模具安装时未测量闭合高度 Doesn't measure the shut height when installing tooling | 2 | 参考冲床仪表,以实际产品尺寸为准 Refer to press instrument, subject to actural dimension | 模具安装后及时检查确认 Check on-time after installing tooling | 2 | 24 | | | | | | | |
| | | 表面擦伤、压印等 Scar or indentation on the surface | 零件外观不合格，顾客满意度下降 Part apperance is not qualified,and customer satisfaction is decreased | 5 | | 模具内部有冲压碎屑 There is scrap inside tooling when stamping | 3 | 作业前用气枪清扫模具内部，并用无纤维布擦拭模具表面 Using an air gun to clean the inside of the mold before operating, and wipe the surface of tools by mold-free fiber cloth | 架模前确认 Confrim before install tooling | 3 | 45 | | | | | | | |
| | | 模面上有废料（跑料） Scrap on tooling | 产品无法使用，模具损坏 Can't do production, the tooling is damaged | 6 | | 吹废料气管没有安装或者气压不足，吹废料动作角度调整不当 Scrap air pipe is not installed or air pressure is not enough, the angle of blowing the scrap is not adjusted well | 2 | 冲压前检查吹废料管的安装状况及空气压力，调整动作角度使吹废料动作正常 Check installation conditions and air pressure of blowing waste pipe before stamping ，adjust the angle of movement to normalize the movement of waste blowing | 班前检查 Check before each shift | 4 | 48 | | | | | | | |
| | | 产品功能边有毛刺超差 Product features have burr out-of-tolerance | 影响产品使用，或降低其功能 Affect the product use, or reduce its function | 6 | | 模具损坏或模具有塌角 Die mold damage or have fallen Angle | 2 | 定期检查模具成型部位，定期维护 Regularly check the molding parts, regular maintenance | 首末检、制程抽检 Check the first and final part,casual check during production | 3 | 45 | | | | | | | |
| | |   1# 尺寸超差 Dimension deviation | 产品装配不良，性能下降 The performance of part is decreased, and it can't be used | 7 | 1 | 1、模具尺寸加工不良；2、模具凸凹模配合间隙有偏差Mold size processing, mold fitting clearance intensive have deviation | 2 | 1、提高模具加工精度，并做好模具尺寸测量；2、定期模具保养1, improve machining precision mould, and make the mould size measurement; 2, regular mold maintenance | 首末检、制程抽检 Check the first and final part,casual check during production | 7 | 98 | SPC控制 SPC control | 程田峰 chen zhenhui | SPC控制 SPC control | 7 | 2 | 7 | 98 |
| | |   7# 尺寸超差 Dimension deviation | 产品装配不良，性能下降 The performance of part is decreased, and it can't be used | 7 | 1 | 1、模具尺寸加工不良；2、模具凸凹模配合间隙有偏差Mold size processing, mold fitting clearance intensive have deviation | 2 | 1、提高模具加工精度，并做好模具尺寸测量；2、定期模具保养1, improve machining precision mould, and make the mould size measurement; 2, regular mold maintenance | 首末检、制程抽检 Check the first and final part,casual check during production | 7 | 98 | SPC控制 SPC control | 程田峰 chen zhenhui | SPC控制 SPC control | 7 | 2 | 7 | 98 |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | S | | | O | 预防 Prevent | 探测 Detect | D | | | | 采取的措施 Actions taken | S | O | D | RPN |
| | |  8# 尺寸超差 Dimension deviation | 产品装配不良，性能下降 The performance of part is decreased, and it can't be used | 7 | 1 | 1、模具尺寸加工不良；2、模具凸凹模配合间隙有偏差Mold size processing, mold fitting clearance intensive have deviation | 2 | 1、提高模具加工精度，并做好模具尺寸测量；2、定期模具保养1, improve machining precision mould, and make the mould size measurement; 2, regular mold maintenance | 首末检、制程抽检 Check the first and final part,casual check during production | 7 | 98 | SPC控制 SPC control | 程田峰 chen zhenhui | SPC控制 SPC control | 7 | 2 | 7 | 98 |
| | |  9# 尺寸超差 Dimension deviation | 产品装配不良，性能下降 The performance of part is decreased, and it can't be used | 7 | 1 | 1、模具尺寸加工不良；2、模具凸凹模配合间隙有偏差Mold size processing, mold fitting clearance intensive have deviation | 2 | 1、提高模具加工精度，并做好模具尺寸测量；2、定期模具保养1, improve machining precision mould, and make the mould size measurement; 2, regular mold maintenance | 首末检、制程抽检 Check the first and final part,casual check during production | 7 | 98 | 检具控制 checking tools control | 程田峰 chen zhenhui | | | | | |
| 4 | 电镀 Plating | 产品外观擦伤、划伤 Product scratch | 产品密封性能下降或失效 Product sealing performance decline or failure | 6 | | 1、清洗栏保护管老化脱落 2、作业方法不当 1. The protection tube of the cleaning bar is aging and falling 2. Improper operation method | 1 | 1、清洗栏定期维护 2、人员培训合格后上岗 3、指导书规定产品放置方法 1. Regular maintenance of the cleaning bar 2. Work after qualified personnel training 3. The instruction book specifies the product | 1、作业人员外观自检 2、品质人员外观巡检 3.品质巡检确认作业方法及点检情况 1. Appearance self-inspection of operators 2. Appearance inspection of quality | 6 | 36 | SPC控制 SPC control | 程田峰 chen zhenhui | SPC控制 SPC control | 7 | 2 | 6 | 84 |
| | | 电镀厚度≥3um 膜厚不达标 The film thickness does not meet the standard | 1、功能面性能下降 Functional performance degradation | 6 | | 1、电镀不到位 1、Inadequate electroplating | 1 | 1、确定膜厚要求 Determine film thickness requirements | 1、每批次电镀产品检测 1、Testing of each batch of electroplated products | 6 | 36 | SPC控制 SPC control | 程田峰 chen zhenhui | SPC控制 SPC control | 7 | 2 | 6 | 84 |
| | | 产品颗粒度不达标 Customer request product granularity is not up to standard | 产品性能下降或失效 Product performance decline or failure | 7 | | 1、电镀条件不当 2、设备异常造成 3、作业不当，导致产品污染 1.Improper electroplating conditions 2. Caused by abnormal equipment 3. Improper operation results in product contamination | 1 | 1、生产前点检设备参数情况 2、每班点检药剂槽浓度 3、设备定期点检 1. Check equipment parameters before production 2. Check the concentration of the reagent tank in each shift | 1、作业人员外观自检 2、品质人员外观巡检 3.品质巡检确认作业方法及点检情况 1. Appearance self-inspection of operators 2. Appearance inspection of quality | 6 | 42 | | | | | | | |
| | |  1# 尺寸超差 Dimension deviation | 产品装配不良，性能下降 The performance of part is decreased, and it can't be used | 7 | 1 | 1、模具尺寸加工不良；2、模具凸凹模配合间隙有偏差Mold size processing, mold fitting clearance intensive have deviation | 2 | 1、提高模具加工精度，并做好模具尺寸测量；2、定期模具保养1, improve machining precision mould, and make the mould size measurement; 2, regular mold maintenance | 首末检、制程抽检 Check the first and final part,casual check during production | 7 | 98 | SPC控制 SPC control | 程田峰 chen zhenhui | SPC控制 SPC control | 7 | 2 | 6 | 84 |
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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | S | | | O | 预防 Prevent | 探测 Detect | D | | | | 采取的措施 Actions taken | S | O | D | RPN |
| 5 | 压装 Press-fitting |  8# 尺寸超差 Dimension deviation | 产品装配不良，性能下降 The performance of part is decreased, and it can't be used | 7 | 1 | 1、模具尺寸加工不良；2、模具凸凹模配合间隙有偏差Mold size processing, mold fitting clearance intensive have deviation | 2 | 1、提高模具加工精度，并做好模具尺寸测量；2、定期模具保养1, improve machining precision mould, and make the mould size measurement; 2, regular mold maintenance | 首末检、制程抽检 Check the first and final part,casual check during production | 7 | 98 | SPC控制 SPC control | 程田峰 chen zhenhui | SPC控制 SPC control | 7 | 2 | 7 | 98 |
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| | |  10# 尺寸超差 Dimension deviation | 产品装配不良，性能下降 The performance of part is decreased, and it can't be used | 7 | 1 | 1、模具尺寸加工不良；2、模具凸凹模配合间隙有偏差Mold size processing, mold fitting clearance intensive have deviation | 2 | 1、提高模具加工精度，并做好模具尺寸测量；2、定期模具保养1, improve machining precision mould, and make the mould size measurement; 2, regular mold maintenance | 首末检、制程抽检 Check the first and final part,casual check during production | 7 | 98 | SPC控制 SPC control | 程田峰 chen zhenhui | SPC控制 SPC control | 7 | 2 | 7 | 98 |
| | |  12# 尺寸超差 Dimension deviation | 产品装配不良，性能下降 The performance of part is decreased, and it can't be used | 7 | 1 | 1、模具尺寸加工不良；2、模具凸凹模配合间隙有偏差Mold size processing, mold fitting clearance intensive have deviation | 2 | 1、提高模具加工精度，并做好模具尺寸测量；2、定期模具保养1, improve machining precision mould, and make the mould size measurement; 2, regular mold maintenance | 首末检、制程抽检 Check the first and final part,casual check during production | 7 | 98 | 检具控制 checking tools control | 程田峰 chen zhenhui | | | | | |
| | |  13# 尺寸超差 Dimension deviation | 产品装配不良，性能下降 The performance of part is decreased, and it can't be used | 7 | 1 | 1、模具尺寸加工不良；2、模具凸凹模配合间隙有偏差Mold size processing, mold fitting clearance intensive have deviation | 2 | 1、提高模具加工精度，并做好模具尺寸测量；2、定期模具保养1, improve machining precision mould, and make the mould size measurement; 2, regular mold maintenance | 首末检、制程抽检 Check the first and final part,casual check during production | 7 | 98 | 检具控制 checking tools control | 程田峰 chen zhenhui | | | | | |
| | | 产品压伤 Product scratch | 产品性能下降或失效 Product performance decline or failure | 6 | | 1、工装装配不良 2、作业方法不当 1. TPoor assembly of tooling 2. Improper operation method | 1 | 1、按压装指导书安装 2、人员培训合格后上岗 3、指导书规定产品放置方法 1. Installation Guide for Press Installation 2. Work after qualified personnel training 3. The instruction book specifies the product | 1、作业人员外观自检 2、品质人员外观巡检 3.品质巡检确认作业方法及点检情况 1. Appearance self-inspection of operators 2. Appearance inspection of quality | 6 | 36 | | | | | | | |
| | | 压装后镀层损坏 Product scratch | 产品性能下降或失效 Product performance decline or failure | 6 | | 1、工装装配不良 2、作业方法不当 1. TPoor assembly of tooling 2. Improper operation method | 1 | 1、按压装指导书安装 2、人员培训合格后上岗 3、指导书规定产品放置方法 1. Installation Guide for Press Installation 2. Work after qualified personnel training 3. The instruction book specifies the product | 1、作业人员外观自检 2、品质人员外观巡检 3.品质巡检确认作业方法及点检情况 1. Appearance self-inspection of operators 2. Appearance inspection of quality | 6 | 36 | | | | | | | |

| 产品过程编号 Process No. | 项目/功能 Item/Function | 潜在失效模式 Potential failure mode | 潜在失效后果 Potential failure effects | 严重度 Severity | 级别 Level | 潜在失效起因/机理 Potential cause/ Mechanism of failure | 频度 Occurrence | 现行过程控制 Current Controls | | 探测度 Detectivity | RPN | 建议措施 Recommended action | 责任及目标完成日期 Responsi-bility & Target completion | 措施结果 Actions results | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | S | | | O | 预防 Prevent | 探测 Detect | D | | | | 采取的措施 Actions taken | S | O | D | RPN |
| | | 产品颗粒度不达标 Customer request product granularity is not up to standard | 产品性能下降或失效 Product performance decline or failure | 7 | | 1、电镀条件不当<br>2、设备异常造成<br>3、作业不当，导致产品污染<br>1.Improper electroplating conditions<br>2. Caused by abnormal equipment<br>3. Improper operation results in product contamination | 1 | 1、生产前点检设备参数情况<br>2、每班点检药剂槽浓度<br>3、设备定期点检<br>1. Check equipment parameters before production<br>2. Check the concentration of the reagent tank in each shift | 1、作业人员外观自检<br>2、品质人员外观巡检<br>3.品质巡检确认作业方法及点检情况<br>1. Appearance self-inspection of operators<br>2. Appearance inspection of quality | 6 | 42 | | | | | | | |
| 6 | 分选/成品检验 Finished product inspection | 错检、漏检Fault detection residual | 不良品混入良品交付 Defective parts are mixed with qualified parts when delivery | 4 | | 人为失误、人员检测技能不到位 Human error and poor staff detection methods | 2 | 不定期抽检 Spot check unscheduled | 上岗前培训考核、加强教育管理Training before taking up their assessment, strengthen the management of education | 7 | 56 | | | | | | | |
| 7 | 包装 Packing | 包装材料不符 Package material is inconsistent | 产品受损 Part is damaged | 3 | | 包装材料拿错 Take wrong package material | 2 | 包装作业指导书 Package operation instruction | 每批出货前检验 Check each batch parts before delivery | 7 | 42 | | | | | | | |
| | | 数量不符 Quantity is inconsistent | 影响交货、客户不满 Affect delivery, customer is not satisfied | 2 | | 点数错误 Account the quantity wrong | 2 | 不定期抽检 Spot check unscheduled | 每批出货前检验 Check each batch parts before delivery | 7 | 28 | | | | | | | |
| | | 标识不符 Mark is inconsistent | 增加找寻时间 Increase the sorting time | 4 | | 标识不正确 Mark is not right | 2 | 不定期抽检 Spot check unscheduled | 每批出货前检验 Check each batch parts before delivery | 7 | 56 | | | | | | | |
| 8 | 产品存放 Product store | 变形 Deformation | 影响使用 Affect to use | 4 | | 堆放不合理，造成倒塌 Doesn't stack well, result in collapse | 2 | 规定堆放高度等要求 Define the requirement of stack height etc | 每批出货前检验 Check each batch parts before delivery | 7 | 56 | | | | | | | |
| | | 表面赃物（灰尘） surface Spoils（dust） | 零件外观不合格，顾客满意度下降 Part apperance is not qualified,and customer satisfaction is decreased | 6 | | 周转箱没有封盖 The pass box without seal-capping | 2 | 不定期抽检 Spot check unscheduled | 上岗前培训考核 Train and qualify before being operator | 2 | 24 | | | | | | | |
| 9 | 出货检验 Outgoing quality control | 漏检 Missing check | 不良品混入良品交付 Bad product with good parts delivery | 4 | | 量具没有校准或超出有效使用期 Gauge without calibration or beyond the effective age | 2 | 量具定期校准并贴校准标签 Gauge calibration and calibration labels on a regular basis | 样件检验量具校准 Sample inspection gage calibration | 4 | 32 | | | | | | | |
| | | | | 4 | | 人为失误 Human error | 2 | 上岗前培训考核 Mount guard training before examination | 样件检查作业员培训记录表 Sample check operator training record | 4 | 32 | | | | | | | |
| | | 检测报告和各项单据漏放 Haven't put Inspection report or other sheet in | 客户无法得到应有信息，造成客户不满 Customer is not satisfied because information is not available | 4 | | 人为失误 Human error | 2 | 不定期抽检 Spot check unscheduled | 每批出货前检验 Check each batch parts before delivery | 6 | 48 | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | S | | | O | 预防 Prevent | 探测 Detect | D | | | | 采取的措施 Actions taken | S | O | D | RPN |
| 10 | 产品出货 Product shipment | 运输车辆没有及时到达 The delivery car not arrives on time | 延误客户交货时间，对客户的生产和物流管理造成影响 Delivery is delayed,which affect customer's production and logistic management | 4 | | 没有提前预约车辆，或者车辆有故障没有提前通知改用其它运输方式 Doesn't book car in advance, or car has trouble without inform in advance. Change into other transportation | 2 | 公司车辆使用管理制度 Manage cars of company by system | 业务每批出货跟踪 Track each batch after delivery | 6 | 48 | | | | | | | |
| | | | | | | 人为失误 Human error | 2 | 上岗前培训考核 Train and qualify before being operator | 培训记录表 Training record chart | 6 | 48 | | | | | | | |
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